THE PROJECTION FORMULA HOLDS WITH A PERFECT MODULE AS THE TWIST

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites one cross-chapter label, recorded in assembly.external_references with its canonical tag as a navigation pointer.

Editorial note. The proof cites no bibliography item inside the counted span.

Allowed standard prerequisites (editorial).

- Perfect complexes are locally represented by finite complexes of finite free modules. Stacks Project, Cohomology of Sheaves, Lemma lemma-perfect-on-locally-ringed (tag 0B8E). Justifies the first reduction of the proof.

Source reference. Original equation 0B53: sources/stacks-chow/tags.txt, source lines 14021–14021. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of ringed spaces. Let $E \in D(\mathcal{O}_X)$ and $K \in D(\mathcal{O}_Y)$. If K is perfect, then*

$$Rf_*E \otimes^{\mathbf{L}}_{\mathcal{O}_Y} K = Rf_*(E \otimes^{\mathbf{L}}_{\mathcal{O}_X} Lf^*K)$$

in $D(\mathcal{O}_Y)$.

Proof. To check (0B53) is an isomorphism we may work locally on Y, i.e., we have to find an open covering $\{V_j \to Y\}$ such that the map restricts to an isomorphism on V_j. By definition of perfect objects, this means we may assume K is represented by a strictly perfect complex of $\mathcal{O}_Y$-modules. Note that, completely generally, the statement is true for $K = K_1 \oplus K_2$, if and only if the statement is true for K_1 and K_2. Hence we may assume K is a finite complex of finite free $\mathcal{O}_Y$-modules. In this case a simple argument involving stupid truncations reduces the statement to the case where K is represented by a finite free $\mathcal{O}_Y$-module. Since the statement is invariant under finite direct summands in the K variable, we conclude it suffices to prove it for $K = \mathcal{O}_Y[n]$ in which case it is trivial. □

Date: stacks-project:cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.